

ECON 21030: Honors Econometrics

Practice Midterm 2

Spring 2023

There are a total of **120 points** on the exam.

1: *[35 points total]* You are interested in the effect of a binary treatment on an outcome of interest. To this end, you collect an iid sample of n individuals and assign them to the treatment or control group with equal probability. Let Y_i denote the outcome of the i th individual. It is assumed that $E[Y^2] < \infty$. Let D_i denote the treatment status of the i th individual, where $D_i = 1$ if the i th individual is treated and $D_i = 0$ if the i th individual is not treated. You assume the following model for Y_i :

$$Y_i = \alpha_i + \beta_i D_i$$

where (α_i, β_i) are independent of D_i . Note that α_i and β_i are random variables, and β_i is the effect of the treatment on the outcome. Furthermore the effect is "heterogeneous" in the sense that β_i differs across individuals.

1. *[5 points]* Explain why it might be reasonable to assume that (α_i, β_i) is independent of D_i .
2. *[10 points]* Consider the linear regression:

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

which you interpret as the best linear predictor of Y_i given D_i . Express α and β in terms of the distribution of (α_i, β_i) . In particular, show that $\beta = E[\beta_i]$

3. *[5 points]* Do you think ε_i is heteroskedastic or homoskedastic? Why?
(*hint:* try writing $\varepsilon_i = \alpha_i - E[\alpha_i] + (\beta_i - E[\beta_i]) D_i$ and investigating $\text{Var} \varepsilon_i | D_i$ for both possible values of D_i .)

4. [5 points] Define $X_i = (1, D_i)$ and define $\hat{\alpha}_n, \hat{\beta}_n$ as the OLS estimators of α and β given our sample. Recall from our usual asymptotic theory that we know:

$$\sqrt{n} \left((\hat{\alpha}_n, \hat{\beta}_n) - (\alpha, \beta)' \right) \xrightarrow{d} N(0, \Omega)$$

Write down an expression for Ω as well as a consistent estimator of Ω . Call this estimator $\hat{\Omega}_n$. You do not need to show that $\hat{\Omega}_n$ is consistent.

5. [10 points] Suppose we are interested in obtaining a confidence interval around $\hat{\beta}_n$, but we do not care about $\hat{\alpha}_n$. Define $r' = (0, 1)'$ and show how to use this vector to derive a confidence interval for $\hat{\beta}_n$ at level $a \in (0, 1)$.

2: [45 points total] Consider an i.i.d. sample $\{Y_i, D_i\}_{i=1}^n$ where Y_i is future earnings and $D_i \in \{0, 1\}$ indicates whether an individual received a scholarship to attend college. Let $Y_i(0)$ and $Y_i(1)$ represent the potential outcomes corresponding to the events $D_i = 0$ and $D_i = 1$, respectively.

1. [10 points] Suppose we write down the structural model:

$$Y_i = \beta_0 + \beta_{1,i} D_i + U_i$$

Where we assume $\mathbb{E}(U_i) = 0$ and define $\beta_0 = E[Y_i(0)]$. Provide an expression for the $\beta_{1,i}$ and U_i in terms of the potential outcomes and their first moments. How would you interpret the random variable $\beta_{1,i}$? Is it identified for any individual i ? Explain.

2. [10 points] Is $\kappa = \mathbb{E}(Y_i|D_i = 1) - \mathbb{E}(Y_i|D_i = 0)$ identified? When is κ equal to ATT , $ATUT$, or ATE ?
3. [10 points] Suppose the scholarship is only given to high-achieving students, who are already more likely to have higher earnings. Will κ overstate or understate the ATT and the $ATUT$?
4. [10 points] Suppose the scholarship is given based on financial need, so that its recipients are more positively affected by D_i than other students are. How do you expect the ATT , $ATUT$, and ATE will compare in magnitude? Provide intuition behind your comparisons.
5. [5 points] Suppose that treatment D_i is randomized so that $\Pr(D_i = 1) = 0.5$. Show that $\kappa = ATE$.

3: [40 points total] Consider the following model of supply and demand:

$$\begin{aligned} Q^D &= \beta_0^D + \beta_1^D P + U^D & (\text{Demand}) \\ Q^S &= \beta_0^S + \beta_1^S P + U^S & (\text{Supply}) \end{aligned}$$

Throughout this exercise, assume that $E(U^D) = E(U^S) = E(U^D U^S) = 0$, and suppose you observe the market in equilibrium (i.e. $Q^D = Q^S$). Your goal is to estimate β_1^D and β_1^S .

1. [10 points] Show that P is endogenous in both equations.
2. [10 points] Suppose you observe a variable Z that affects only the supply, so that:

$$\begin{aligned} Q^D &= \beta_0^D + \beta_1^D P + U^D \\ Q^S &= \beta_0^S + \beta_1^S P + \beta_2^S Z + U^S \end{aligned}$$

Assume that $E(ZU^D) = E(ZU^S) = 0$ and $\beta_2^S \neq 0$. Show that Z satisfies both instrument exogeneity and instrument relevance. Do these properties hold if Z also affects demand?

3. [10 points] Using the setup from (b), express both P and Q as linear functions of Z . Then, assuming that Z is a valid instrument, show how to consistently estimate β_1^D from data $\{Q_i, P_i, Z_i\}_{i=1}^n$.
4. [10 points] Now, suppose that consumers in the market face a tax τ , so that:

$$\begin{aligned} Q^D &= \beta_0^D + \beta_1^D (P + \tau) + U^D \\ Q^S &= \beta_0^S + \beta_1^S P + U^S \end{aligned}$$

Assume $\tau \perp (U^D, U^S)$. Given $\{Q_i, P_i, \tau_i\}$, show how to consistently estimate β_1^D and β_1^S . How would you interpret the coefficients β_1^D and β_1^S ? Justify your response.